

Oracle EPM Predictive Cash Forecasting

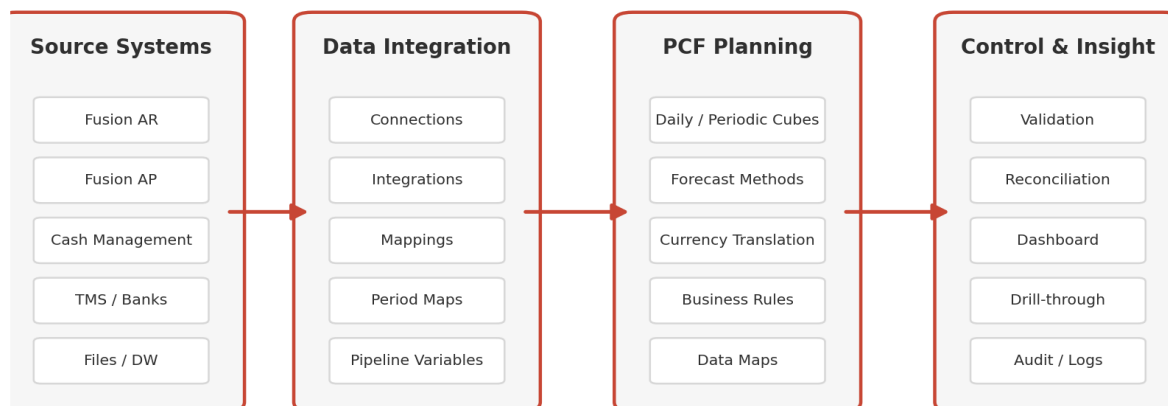
DATA INTEGRATION PIPELINES

Advanced Lab Guide & Technical Reference

Purpose

Product design documentation | Implementation guide | Training material | Technical reference | Troubleshooting handbook

Predictive Cash Forecasting - Data Integration Pipeline Architecture



Pipeline orchestration provides sequencing, dependency control, runtime variables, status visibility, restartability and auditability.

Audience	Recommended Background
Advanced Oracle EPM / Planning students	Data Integration, Planning, cash forecasting, mappings, business rules
Implementation consultants	Cloud EPM administration, Fusion ERP concepts, automation
Treasury / FP&A solution architects	Liquidity, reconciliation, controls, operational runbooks

1. Executive Overview

Why pipelines matter in Predictive Cash Forecasting

A Data Integration pipeline is the orchestration layer that turns source transactions and balances into a repeatable, controlled cash-forecast refresh. It coordinates extraction, import, mapping, validation, currency translation, calculation, reporting data maps, reconciliation and operational status as one managed process.

Current Oracle position

Cloud EPM Pipeline orchestrates a series of jobs as one process, providing control and visibility across preprocessing, data loading and post-processing. Predictive Cash Forecasting now includes pre-seeded Data Integration pipelines for Fusion Cloud ERP AR, AP and Cash Management integrations.

What an advanced implementation must solve

- Deterministic sequencing and job dependencies
- Runtime variables for period, scenario, version, entity and source scope
- Source-to-target reconciliation and fail-fast validation gates
- Restartability without duplicating previously loaded data
- Currency translation and reporting-cube publication
- Custom dimensions without losing Oracle-delivered upgrades
- Auditability, security, SLA monitoring and incident evidence

End-to-End PCF Pipeline Process

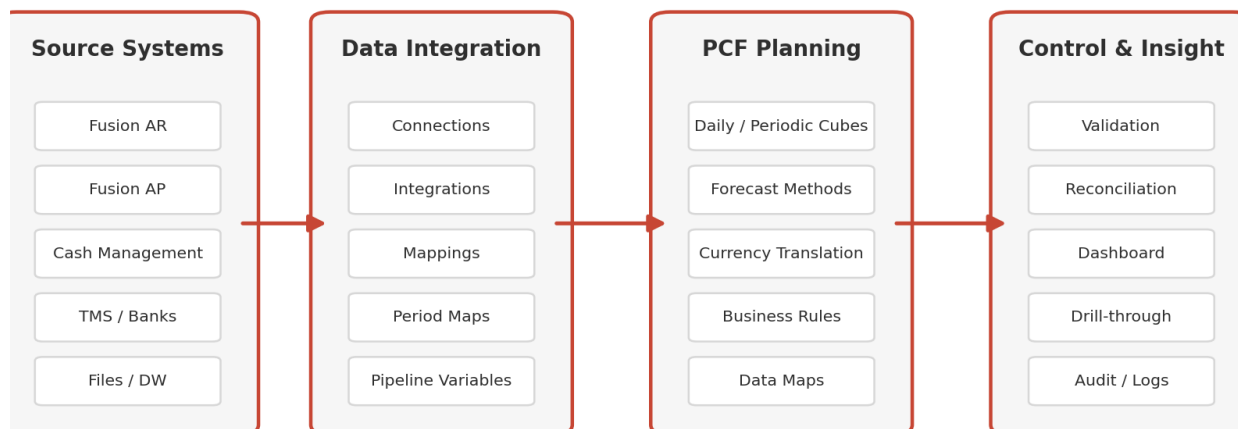
Advanced-stage orchestration pattern with control gates



2. Reference Architecture

Layered pipeline architecture for PCF

Predictive Cash Forecasting - Data Integration Pipeline Architecture



Pipeline orchestration provides sequencing, dependency control, runtime variables, status visibility, restartability and auditability.

Layer responsibilities

Layer	Primary responsibility	Typical controls
Source	Fusion ERP, banks, TMS, files, DW	Completeness, cutoff, source timestamp
Data Integration	Connections, integrations, mappings, period maps	Row counts, rejects, mapping coverage
Pipeline	Sequence and dependency orchestration	Variables, status, restart point, error branch
PCF Planning	Forecast cubes, rules, methods, translation	Calculation status, FX rates, opening/closing cash
Reporting	ASO reporting, dashboards, drill-through	Reconciliation, KPI freshness, traceability

Design principle

Keep source extraction logic, mapping logic, calculation logic and orchestration logic separate. A pipeline should coordinate jobs; it should not become the only place where business logic is hidden.

3. Oracle Pipeline Concepts

Jobs, variables, dependencies and cross-instance orchestration

Concept	Design use in PCF	Advanced note
Pipeline	Single controlled run for full cash-refresh cycle	Use a business-oriented name and versioning standard
Stage / Job	One executable unit	Keep stages idempotent where possible
Variable	Parameter passed at runtime	Use for StartPeriod, EndPeriod, Scenario, Version, Entity
Dependency	Run next job only after required predecessor	Add validation before destructive or publish steps
Cross-instance job	Coordinate jobs across EPM instances	Useful where source or reporting workloads are separated
Status / log	Run evidence and diagnostics	Capture run ID in reconciliation evidence

Recommended naming convention

Artifact	Example
Pipeline	PCF_CASH_REFRESH_DAILY_V1
Integration	PCF_ERP_AR_OPEN_ITEMS
Location	PCF_ERP_AR_US
Data Rule	PCF_AR_LOAD
Business Rule	PCF_Currency_Translate
Data Map	Daily Cash to Reporting

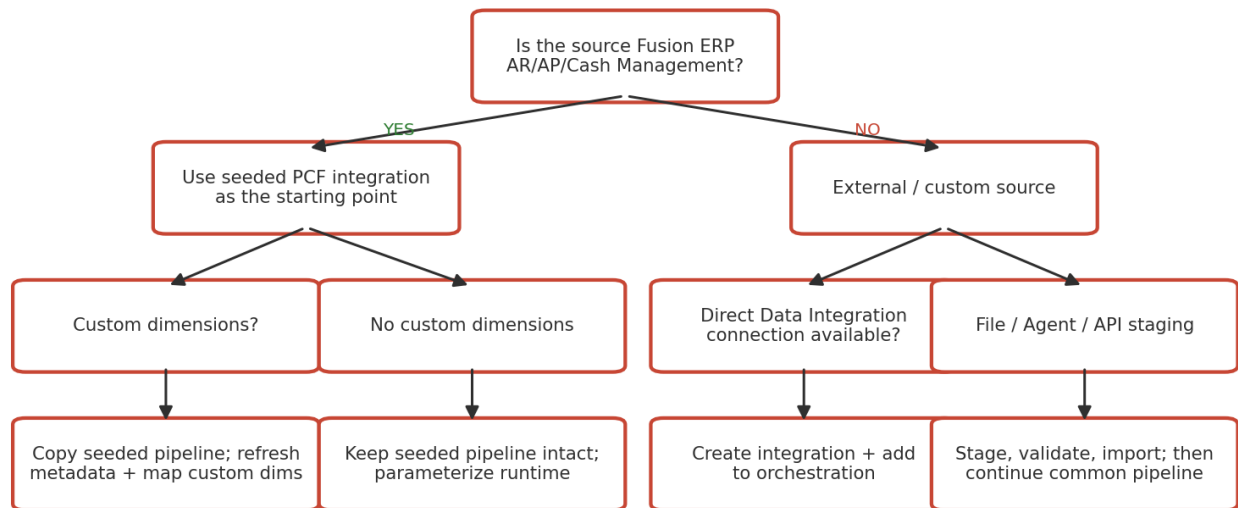
Upgrade-safe rule

For Oracle-provided PCF ERP pipelines, copy the out-of-box pipeline before customization. This is especially important when adding custom dimensions, because future Oracle updates can replace delivered artifacts.

4. Pipeline Design Decision Tree

Choose the right orchestration pattern

Pipeline Design Decision Tree



Decision outcomes

- Seeded Fusion ERP integration: preserve delivered logic and parameterize runtime.
- Seeded + custom dimensions: copy pipeline, refresh metadata, map new dimensions and test upgrade regression.
- External source with native connection: build source integration and attach it to the common orchestration.
- File/agent/API source: establish a controlled staging contract, validate the payload, then enter the common pipeline.

5. Fusion ERP Seeded PCF Pipelines

AR, AP and Cash Management orchestration

Oracle PCF integration with Fusion Cloud ERP provides ready-to-use integrations for Accounts Receivable, Accounts Payable and Cash Management and pre-seeded Data Integration pipelines. The integration also supports drill-through to ERP analytical views and detailed transactions when security and configuration are aligned.

Source	Cash forecasting purpose	Typical forecast contribution
Receivables	Open customer invoices and collection timing	Customer receipts / operating inflow
Payables	Open supplier invoices and payment timing	Supplier payments / operating outflow
Cash Management	Bank balances and financing / cash transactions	Opening cash, booked/available balances, financing activity

Implementation sequence

- Complete Cloud ERP Predictive Cash Forecasting enablement and extraction options.
- Configure Cloud EPM connection to ERP and SSO/security alignment.
- Create a Planning application of type Cash Forecasting and enable required PCF features.
- Validate seeded integrations independently before running the combined pipeline.
- Enable drill-through only after entity/data-access security matches across ERP and EPM.

Advanced control

Do not treat a successful HTTP/extract response as a successful business load. Reconcile source amount/count, staging amount/count and target amount/count before publishing the forecast.

6. Pipeline Configuration Mockup

Oracle-inspired orchestration UI concept

Data Integration / Pipelines / PCF_CASH_REFRESH

Pipeline Variables

StartPeriod
W01

EndPeriod
W13

Scenario
Forecast

Version
Working

Entity
Total_Entity

Job Sequence

01 ERP_AR	On Success -> Next Job
02 ERP_AP	On Success -> Next Job
03 CASH_MGMT	On Success -> Next Job
04 EXT_TMS	On Success -> Next Job
05 VALIDATE	On Success -> Next Job
06 FX_TRANSLATE	On Success -> Next Job
07 DATA_MAP	On Success -> Next Job
08 CALCULATE	On Success -> Next Job
09 RECONCILE	On Success -> Next Job

Publish run status

Run Pipeline

Required runtime parameters

Variable	Example	Validation
StartPeriod	W01	Must exist in enabled forecast calendar
EndPeriod	W13	Must be >= StartPeriod
Scenario	Forecast	Allowed scenario only
Version	Working	Writable target version
Entity	Total_Entity	Valid entity scope / security
RunMode	FULL / RESTART	Controls clear-and-reload vs restart behavior

7. End-to-End Job Sequence

A production-grade PCF cash-refresh pipeline

End-to-End PCF Pipeline Process

Advanced-stage orchestration pattern with control gates



#	Stage	Job	Gate / expected result
1	Initialize	Validate variables + calendar	All runtime parameters valid
2	Extract	ERP AR / AP / Cash + external feeds	Source extraction complete
3	Import	Load staging / workbench	No file/schema failures
4	Map & Validate	Apply mappings / transformations	Unmapped = 0 or approved threshold
5	Translate	Run currency translation	Rates available; translation complete
6	Publish	Run Data Maps to reporting	Target intersections updated
7	Calculate	Run PCF rules / balances	No calc errors
8	Reconcile	Compare source-to-PCF totals	Difference within tolerance
9	Publish Status	Dashboard / notification / evidence	Run = Success or Controlled Warning

8. Pipeline Variables and Parameterization

Design for repeatability, not hard-coded periods

Parameter	Why it matters	Example
ForecastStart	Controls active rolling horizon	2026-W35
ForecastEnd	Limits extract and calculation scope	2026-W47
ActualCutoff	Separates actual from forecast	2026-W34
Scenario	Supports Base / Stress / Management Action	Base
Version	Supports controlled working / approved states	Working
EntityScope	Reduces runtime and controls ownership	US_Region
SourceMode	ERP_ONLY / ALL_SOURCES	ALL_SOURCES
RestartFrom	Controlled restart after failure	FX_TRANSLATE

Validation pattern

- Fail immediately when a required variable is blank.
- Validate that period variables map to enabled PCF forecast ranges.
- Validate the entity against both metadata and user access.
- Persist runtime variables in the run audit so the exact forecast refresh can be reconstructed.

9. Custom Dimensions in Seeded Pipelines

Safely extending Oracle-delivered integration

Predictive Cash Forecasting can be extended with additional dimensions. Oracle documentation notes that delivered ERP pipelines do not automatically pick up newly enabled custom dimensions. The implementation must refresh metadata, map source dimensions to target dimensions, and optionally add transformation expressions.

Step	Action	Example
1	Copy delivered pipeline	PCF_ERP_Seeded -> PCF_ERP_Custom_V1
2	Enable custom dimension	TreasuryRegion
3	Refresh target metadata	Dimension becomes available to Data Integration
4	Map source value	BusinessUnit -> TreasuryRegion
5	Load custom dimension metadata separately	US_BU -> NorthAmerica
6	Regression test all seeded source flows	AR, AP, Cash Management
7	Document ownership and upgrade procedure	Never modify delivered pipeline directly

Oracle recommendation

Make a copy of out-of-box pipelines before making modifications. This prevents future updates from overwriting your changes.

10. External Sources Inside the Pipeline

TMS, banks, files and data warehouses

A mature PCF implementation frequently blends Fusion ERP with external liquidity data such as TMS debt schedules, bank statements, investment maturities, payroll, tax schedules or legacy ERP feeds. These should enter the same orchestration only after passing source-specific controls.

Data Flow Through the PCF Pipeline

Source lineage remains traceable across integration, calculation and reporting layers



Source	Example feed	Pipeline control
TMS	Debt draw/repayment schedule	As-of timestamp + facility ID validation
Bank	Opening available balance	Account ownership + duplicate statement check
File	Payroll / tax cash schedule	Filename, schema, checksum, record count
Data warehouse	Historical actual cash	Period completeness + entity/currency coverage

11. Currency Translation and Data Maps

Do not publish stale local-currency values

Oracle recommends running pipeline-controlled currency translation when required. Translation is included in provided ERP pipelines, and may also be necessary after manual adjustments or for non-ERP sources. After translation, Data Maps push converted data from BSO cubes to the ASO reporting cube, including the supplied Daily Cash to Reporting and Periodic Monthly Cash to Reporting maps where applicable.

Control point	Test	Failure action
FX rates	All required entity/currency/period rates exist	Stop before translation
Translation	Rule completed with no errors	Restart from translation after correction
Data Map	Mapped intersections match target metadata	Refresh metadata / mapping
ASO freshness	Reporting timestamp >= current run	Re-run data map only if upstream is valid

Restart design

Separate translation and reporting publication into distinct stages. This allows a failed ASO push to restart without re-extracting ERP transactions.

12. Validation Gates and Reconciliation

Convert technical success into business confidence

Gate	Measure	Example threshold
Source completeness	Expected source files / extracts received	100%
Mapping coverage	Rows mapped to valid target members	100% or approved exceptions
Reject rate	Rejected rows / imported rows	0% target
Amount reconciliation	Source total - target loaded total	\$0 or defined tolerance
Cash roll-forward	Opening + net movement - closing	\$0
Reporting freshness	Latest reporting load timestamp	Current run ID

Recommended reconciliation evidence

- Pipeline Run ID and runtime parameters
- Source row count and amount by source
- Imported/rejected row counts
- Mapped amount by cash-flow line item
- Currency translation status
- PCF closing cash and buffer result
- Final ASO reporting timestamp
- Approver / reviewer and exception comments

13. Advanced Lab - Build PCF_CASH_REFRESH

Scenario and target architecture

Lab scenario

A global treasury team uses Fusion ERP AR/AP/Cash Management plus a TMS feed. The target is a 13-week PCF rolling forecast. All source loads must complete before translation, calculation and reporting publication.

Input	Value
Opening cash	\$12.4m
Forecast horizon	W01-W13
Minimum cash buffer	\$10.0m
Scenario / Version	Forecast / Working
Entity scope	US_Region
Reporting currency	USD

Lab objective

- Create runtime variables and validation logic.
- Orchestrate four source integrations.
- Insert mapping and reconciliation gates.
- Run currency translation and reporting Data Maps.
- Calculate closing cash and liquidity buffer.
- Inject a controlled mapping failure and perform a restart.

14. Lab Step 1 - Source Integration Design

Define sources and target cash line items

Source	Source total	Target PCF line item	Expected sign
Fusion AR	\$8.6m	Customer Receipts	+
Fusion AP	\$5.2m	Supplier Payments	-
Payroll file	\$1.8m	Payroll	-
TMS investments	\$1.5m	Investment Purchases	-
TMS financing	\$2.0m	Debt Drawdown	+
Other operating	\$0.4m	Other Operating Outflow	-

Expected net movement = $8.6 - 5.2 - 1.8 - 1.5 + 2.0 - 0.4 = +1.7\text{m}$.

Expected closing cash = $12.4 + 1.7 = 14.1\text{m}$. Expected cash over minimum buffer = $14.1 - 10.0 = 4.1\text{m}$.

Control result

The pipeline should not publish a closing cash balance unless every source amount reconciles to the target and the roll-forward difference is zero.

15. Lab Step 2 - Create Pipeline Variables

Parameterize the run

Variable	Lab value	Required?
StartPeriod	W01	Yes
EndPeriod	W13	Yes
Scenario	Forecast	Yes
Version	Working	Yes
Entity	US_Region	Yes
RunMode	FULL	Yes
RestartFrom		No

Validation pseudologic

Validation	Expected behavior
Blank StartPeriod	Stop pipeline with ERROR
EndPeriod before StartPeriod	Stop pipeline with ERROR
Invalid Entity	Stop before source extract
Scenario not writable	Stop before import
RunMode=RESTART and RestartFrom blank	Stop and request restart stage

16. Lab Step 3 - Orchestrate and Run

Execute the full refresh

Sequence	Job	Lab result
01	ERP_AR_LOAD	Success - \$8.6m
02	ERP_AP_LOAD	Success - \$5.2m
03	CASH_MGMT_LOAD	Success - opening \$12.4m
04	TMS_LOAD	Success - investing \$1.5m / financing \$2.0m
05	PAYROLL_FILE_LOAD	Success - \$1.8m
06	VALIDATE_MAPPINGS	Success - 100% mapped
07	FX_TRANSLATE	Success
08	DATA_MAP_REPORTING	Success
09	PCF_CALCULATE	Success - closing \$14.1m
10	RECONCILE	Success - difference \$0

Expected dashboard

Pipeline SUCCESS; Closing Cash \$14.1m; Cash Over Buffer \$4.1m; Rejects 0; Reconciliation Difference \$0.

17. Lab Step 4 - Failure Injection and Restart

Troubleshoot without rerunning everything

Injected defect	Observed message	Corrective action
Supplier SUP-449 missing mapping	VALIDATE_MAPPINGS failed: 3 rows unmapped	Add explicit/wildcard mapping; validate staging
Restart mode	RESTART from VALIDATE	Do not repeat successful ERP extracts
FX rate missing for EUR entity	FX_TRANSLATE failed for 2026-W08	Load rate; restart from FX_TRANSLATE
ASO map blocked by metadata mismatch	DATA_MAP failed: invalid member	Refresh metadata / target mapping; restart Data Map

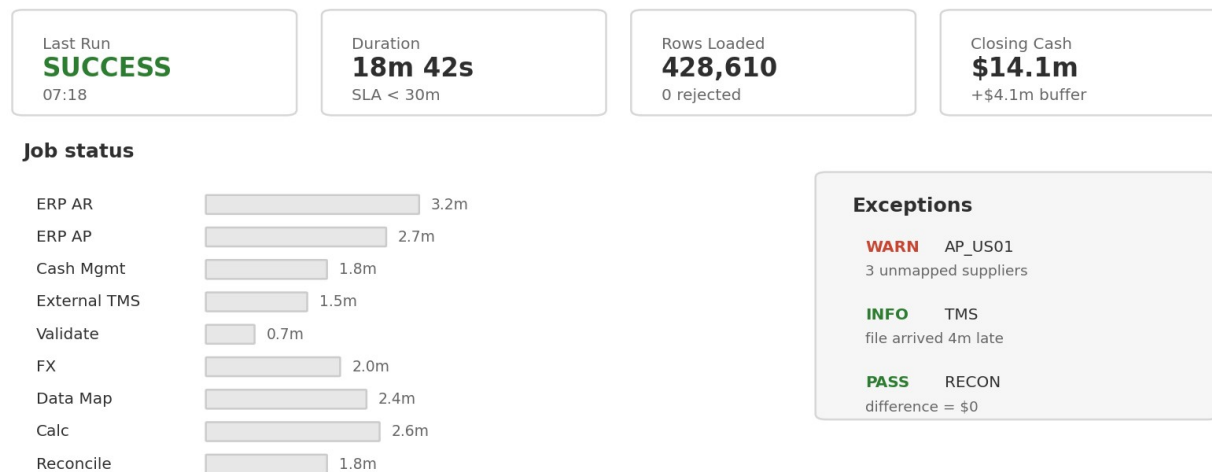
Advanced restart rule

- Restart from the earliest stage whose output is invalid.
- Do not restart source extracts if staged data is complete and immutable.
- If a mapping rule changes, rerun mapping/import for impacted data before calculation.
- If only the reporting Data Map fails, keep the BSO calculation result and republish after correction.

18. Monitoring Dashboard Mockup

Operational visibility for treasury and EPM administrators

Pipeline Operations Cockpit - Mockup



Recommended KPIs

KPI	Purpose
Pipeline success rate	Reliability over rolling period
Median / P95 duration	Capacity and SLA trend
Rows rejected	Data-quality indicator
Unmapped amount	Business impact of mapping defects
Reconciliation difference	Financial control
Forecast freshness	Time since last successful publish
Restart count	Operational stability indicator

19. Security, Roles and Segregation of Duties

Control who can configure, run, approve and drill

Role	Permitted activity	Control
Service Administrator	Connections, pipeline design, security	Privileged access review
Data Integration Admin	Integrations, mappings, pipeline maintenance	No business forecast approval
Cash Manager	Run/review forecast; scenarios	Entity-level data access
Controller / Reviewer	Reconciliation and approval	Read-only source configuration
Operator	Execute approved pipeline	No mapping/config changes

Fusion drill-through

For ERP drill-through, entity/data security must align between Cloud EPM and Fusion Cloud ERP. SSO and ERP job/data-access roles should be configured so users see only authorized transactions.

20. Automation and Scheduling

From manual run to production service

Frequency	Recommended pipeline pattern	Example
Daily	Incremental / rolling refresh	06:30 local treasury cutoff
Weekly	Full periodic refresh + variance snapshot	Monday after ERP close window
On demand	Management action / stress scenario refresh	Triggered after material assumption change
Recovery	RESTART pipeline from controlled stage	After mapping or FX correction

Production controls

- No overlapping run for the same scenario/version/entity scope.
- Lock or serialize destructive clear/load stages.
- Record run ID, start/end time, user/service account and variable values.
- Notify only after reconciliation gate completes.
- Escalate when duration exceeds SLA or the last successful run is stale.

21. Log and Error Examples

Troubleshooting handbook

Severity	Sample log	Interpretation / action
ERROR	PCF-PIPE-101 Required variable StartPeriod is blank	Correct invocation; rerun from Initialize
ERROR	DI-MAP-204 Member Supplier_449 has no target mapping	Create mapping; rerun mapping/import
ERROR	PCF-FX-310 Missing USD/EUR rate for W08	Load FX rate; restart translation
ERROR	DM-402 Invalid member TreasuryRegion_East	Refresh metadata and correct custom-dimension map
WARN	PCF-SRC-120 TMS file received 12 minutes after SLA	Run allowed if treasury approves freshness exception
ERROR	PCF-REC-501 Source/target difference = 0.23m	Do not publish; reconcile missing/duplicate rows
INFO	PCF-RUN-000 Pipeline completed SUCCESS; RunId=PCF2608260718	Archive evidence and publish dashboard

Fast diagnostic sequence

- Read first failed job, not the final pipeline status.
- Confirm whether failure occurred before or after data import.
- Compare source/staging/target counts and amounts.
- Check mappings and metadata before rerunning calculations.
- Use the narrowest safe restart point.
- Reconcile again after recovery; never bypass the final financial control.

22. Performance and Scalability

Designing pipelines for larger transaction volumes

Technique	Benefit	Caution
Scope variables	Restrict entities/periods	Avoid partial reporting without clear status
Parallel independent extracts	Reduce elapsed time	Only parallelize jobs with no dependency
Incremental load	Reduce data movement	Requires reliable source change/cutoff logic
Separate publish stage	Fast recovery from reporting failure	Must preserve BSO run state
Mapping hygiene	Faster validation / fewer rejects	Review wildcard rules for unintended matches
Archive logs/evidence	Keeps operations manageable	Retain according to audit policy

Optimization priority

First remove unnecessary source volume and failed mappings; then optimize job parallelism. Technical concurrency cannot compensate for poor data contracts.

23. Product Design Specification

Functional requirements for a pipeline workbench

Requirement	Design expectation
Pipeline canvas	Ordered stages with dependencies and status
Variable panel	Typed runtime parameters with validation
Run history	Search by run ID, date, user, status, entity
Exception panel	Unmapped/rejected rows with business amount impact
Restart action	Restart from authorized safe stage
Reconciliation card	Source/stage/target/count/amount comparisons
Drill-through	From PCF amount to supporting source detail where available
Audit	Configuration changes, run initiator, approvals and comments

UX behavior

- Use red only for failure/attention, not as decorative noise.
- Show the first failed dependency prominently.
- Display business impact (\$ amount) next to technical error counts.
- Disable Publish/Approve while reconciliation is outside tolerance.

24. Implementation Checklist

Advanced-stage go-live readiness

- ☐ Source inventory and ownership confirmed
- ☐ ERP seeded integrations validated separately
- ☐ External-source data contracts approved
- ☐ Pipeline copied before customizing Oracle-delivered artifacts
- ☐ Runtime variables documented and validated
- ☐ Custom dimensions refreshed and mapped
- ☐ Currency translation stage tested
- ☐ Reporting Data Maps validated
- ☐ Source-to-target reconciliation automated
- ☐ Restart points tested with injected failures
- ☐ SSO and entity-level security aligned
- ☐ Monitoring/SLA/notification design approved
- ☐ Operating runbook and support ownership signed off
- ☐ Regression pack retained for quarterly updates

Go-live criterion

A pipeline is production-ready only when the team can prove the correct data was loaded, the forecast was calculated and published, failures can be restarted safely, and the final cash position reconciles to source evidence.

25. Technical Reference and Key Oracle Notes

Current capability anchors for this guide

Topic	Reference point
Pipeline	Orchestrates a series of EPM jobs as one process and supports cross-instance orchestration.
PCF + Fusion ERP	Pre-seeded Data Integration pipelines are available for AR, AP and Cash Management integration.
Custom dimensions	Delivered ERP pipelines require manual refresh/mapping for additional dimensions; Oracle recommends copying delivered pipelines before modification.
Currency translation	Provided ERP pipeline includes translation; pipeline-based translation may also be needed after adjustments or for non-ERP sources.
Reporting Data Maps	PCF administration documentation identifies Daily Cash to Reporting and Periodic Monthly Cash to Reporting publication maps.
Forecast cycles	PCF supports daily, weekly and monthly rolling forecasts.

Suggested Oracle reading

- Using the Pipeline - Oracle Fusion Cloud EPM Data Integration
- Administering Predictive Cash Forecasting
- Predictive Cash Forecasting: Integration with Oracle Cloud ERP
- Configuring Custom Dimensions in Data Integration
- Working with Predictive Cash Forecasting

Training note

UI screenshots in this guide are Oracle-inspired training mockups, not literal copies of Oracle proprietary product screens. Verify menu names and release-specific behavior in your own environment before production use.